

Medvale Internal Medicine

Chapter 3D: Colon, Rectum, Lower GI Bleeding, and Mesenteric Emergencies

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Why this chapter matters clinically

Colon and rectal disorders sit at the intersection of outpatient prevention, inpatient triage, emergency surgery, oncology, vascular medicine, and gastroenterology. A patient with painless hematochezia may have diverticular hemorrhage, angiodysplasia, brisk upper gastrointestinal bleeding, colorectal cancer, or ischemic colitis. A patient with abdominal distention may have mechanical obstruction, pseudo-obstruction, ileus, toxic megacolon, volvulus, or severe constipation. A patient with abdominal pain out of proportion to the exam may have acute mesenteric ischemia, which can progress from reversible mucosal injury to transmural infarction and septic shock within hours. Because of that overlap, the safest approach is not memorizing isolated diagnoses, but building a triage map that links symptoms, risk factors, imaging, endoscopy, resuscitation, and surgical triggers.

This segment covers colorectal cancer and polyps; diverticulosis and diverticulitis; lower gastrointestinal bleeding; angiodysplasia; acute and chronic mesenteric ischemia; Ogilvie syndrome; colonic volvulus; colorectal obstruction patterns; and practical management pathways. It intentionally repeats several numeric anchors because graph RAG performs better when the same relation appears in more than one context, such as age thresholds, colonoscopy intervals, bleeding localization clues, surgical warning signs, and ischemia mortality risk.

Core illness scripts

Syndrome	Classic script	Do-not-miss danger
Colorectal cancer	Age over 45, positive stool test, iron deficiency anemia, change in bowel habits, weight loss, or obstructive symptoms; right-sided tumors may bleed silently and left-sided tumors more often narrow the stool caliber.	Obstruction, perforation, metastatic disease, synchronous lesions, and missed diagnostic colonoscopy after a positive screening test.
Adenomatous polyps	Often asymptomatic; risk rises with larger size, villous histology, high-grade dysplasia, sessile morphology, and number of adenomas.	Failure to remove or survey high-risk adenomas allows progression along the adenoma-carcinoma sequence.
Diverticulosis	Usually incidental; may cause painless, large-volume hematochezia, especially in older adults.	Recurrent or severe hemorrhage requiring transfusion, endoscopic therapy, angiographic embolization, or surgery.
Diverticulitis	Left lower quadrant pain, fever, leukocytosis, bowel habit change, and localized tenderness; CT is the usual diagnostic test.	Abscess, perforation, fistula, obstruction, sepsis, and misdiagnosed colon cancer.
Angiodysplasia	Older adult with intermittent painless lower GI bleeding; lesions are vascular ectasias, often right-sided.	Recurrent bleeding, anemia, and missed association with aortic stenosis or chronic kidney disease.
Acute mesenteric ischemia	Sudden severe abdominal pain out of proportion to exam, atrial fibrillation or vascular disease, metabolic acidosis, lactate elevation late, and rapid deterioration.	Bowel infarction, peritonitis, septic shock, and death if revascularization or surgery is delayed.

Syndrome	Classic script	Do-not-miss danger
Ogilvie syndrome	Massive colonic dilation without mechanical obstruction in a hospitalized, postoperative, septic, immobilized, or medication-exposed patient.	Cecal perforation risk when dilation is severe or persistent, especially with cecal diameter around or above 10-12 cm.
Sigmoid volvulus	Older or institutionalized patient with constipation, abdominal distention, pain, and coffee-bean/omega-loop imaging pattern.	Closed-loop ischemia, perforation, recurrence after endoscopic detorsion.

Diagnostic geography: right colon, left colon, rectum, and small bowel

The colon is not a uniform tube. Right-sided lesions have a wider lumen and liquid stool, so they may grow large before causing obstruction. They often present as occult blood loss, iron deficiency anemia, melena-like dark stool, fatigue, or vague abdominal discomfort. Left-sided lesions encounter a narrower lumen and more formed stool, so they are more likely to cause a change in bowel habits, constipation alternating with diarrhea, pencil-thin stools, cramping, and overt hematochezia. Rectal lesions may produce tenesmus, urgency, incomplete evacuation, visible blood, or a palpable mass on digital rectal examination. Small bowel bleeding is less common but can present as either melena or hematochezia and is considered after upper endoscopy and colonoscopy fail to find a source.

The ligament of Treitz is the classic anatomic divider: bleeding above it is upper GI bleeding; bleeding below it is lower GI bleeding. This division is useful but imperfect. Massive upper GI bleeding may move through the intestine so quickly that it appears as bright red hematochezia, and small bowel sources may behave like either upper or lower bleeding depending on rate and location. Therefore, hemodynamic status has priority over anatomic assumptions. Shock, orthostasis, syncope, chest pain, dyspnea, or ongoing large-volume bleeding should trigger resuscitation and consideration of an upper source even when the stool is red.

Numeric anchor

In average-risk adults, colorectal cancer screening begins at age 45 and continues through age 75. Ages 76-85 require individualized decisions, and routine screening is generally stopped after age 85. A positive stool-based test is not the endpoint; it requires diagnostic colonoscopy.

Approach to lower gastrointestinal bleeding

Lower GI bleeding is a syndrome, not a diagnosis. It can be trivial hemorrhoidal spotting, life-threatening diverticular hemorrhage, ischemic colitis, colorectal cancer, post-polypectomy bleeding, inflammatory bowel disease, or small bowel hemorrhage. The first decision is whether the patient is unstable. The second is whether the pattern truly localizes below the ligament of Treitz. The third is whether bleeding is active enough to require urgent localization by colonoscopy, CT angiography, tagged red blood cell scan, or angiography.

Patient with hematochezia, melena, or occult blood

Assess airway, mental status, perfusion, orthostasis, anticoagulants, antiplatelets, NSAIDs, cirrhosis, aortic graft history, and cardiac disease.

THEN

Unstable or brisk bleeding?

Two large-bore IVs, type and cross, CBC, CMP, PT/INR, PTT, platelets, lactate if shock, transfuse/resuscitate, reverse severe coagulopathy when appropriate.

THEN

Consider upper source if unstable hematochezia

Upper endoscopy is often prioritized when melena, hematemesis, cirrhosis/varices, high BUN-to-creatinine ratio, or shock with red blood is present.

THEN**Stable suspected lower source**

Bowel preparation followed by colonoscopy, with therapeutic clipping, coagulation, injection, or banding when a culprit lesion is identified.

THEN**Ongoing active bleeding or cannot prep?**

CT angiography can localize active bleeding; angiography can embolize; surgery is reserved for persistent severe bleeding or failure of endoscopic/angiographic control.

Finding	Interpretation	Practical implication
Hematemesis	Usually upper GI source proximal to ligament of Treitz.	Resuscitate and perform upper endoscopy after stabilization.
Coffee-ground emesis	Upper GI bleeding at a slower rate or partially digested blood.	Still treat as upper GI bleeding until proven otherwise.
Melena	Often upper GI source, but jejunal, ileal, or right colonic sources are possible.	Upper endoscopy first in most cases; colonoscopy if EGD is negative or suspicion remains.
Hematochezia	Usually left colon, rectum, or brisk lower source; 5-10% can be massive upper GI bleeding.	Assess hemodynamics before assuming lower source.
Occult blood + iron deficiency anemia	Chronic blood loss until proven otherwise, especially colorectal neoplasia in adults.	Bidirectional endoscopy is commonly needed, depending on age and symptoms.
High BUN/creatinine ratio	Digested blood protein can raise BUN in upper GI bleeding if renal function is not the explanation.	Supports, but does not prove, an upper source.

Initial management of GI bleeding

Resuscitation comes before localization. Two large-bore peripheral IVs are preferred over a single small IV. Type and cross-match blood if bleeding is moderate or severe. Hemoglobin may appear falsely reassuring early because whole blood is lost before plasma redistribution lowers the measured concentration. Trend hemoglobin every 4-8 hours in active bleeding, but treat the patient rather than the number. Tachycardia, orthostasis, hypotension, syncope, cool extremities, altered mental status, reduced urine output, and chest pain are more urgent than a single laboratory value.

Transfusion thresholds depend on clinical context. A restrictive strategy around hemoglobin 7 g/dL is often used for stable patients, while higher targets may be used in active shock, severe cardiopulmonary disease, acute coronary syndrome, or ongoing hemorrhage. Platelet dysfunction from antiplatelet agents, supratherapeutic anticoagulation, cirrhosis-associated coagulopathy, and renal failure can all change the risk calculus. Reversal of anticoagulation is reserved for severe, unstable, or life-threatening bleeding because thrombosis risk also matters.

Common causes of lower GI bleeding

Cause	Typical patient/pattern	Diagnosis	Management concept
Diverticular bleeding	Older adult; painless, abrupt, sometimes large-volume hematochezia. Often stops spontaneously.	Colonoscopy after prep if stable; CT angiography if active severe bleeding.	Endoscopic hemostasis if stigmata found; embolization for active severe bleeding; surgery rarely.
Angiodysplasia	Age over 60, chronic kidney disease, aortic stenosis association; intermittent painless bleeding or iron deficiency.	Colonoscopy; sometimes capsule endoscopy or deep enteroscopy if small bowel suspected.	Endoscopic coagulation; treat anemia; recurrent cases may need repeat therapy.

Cause	Typical patient/pattern	Diagnosis	Management concept
Ischemic colitis	Crampy pain followed by bloody diarrhea in older vascular patient, low-flow state, or hypercoagulability.	CT and colonoscopy when safe; avoid colonoscopy in peritonitis.	Supportive care, IV fluids, bowel rest, antibiotics if moderate/severe, surgery for gangrene/perforation.
Colorectal cancer/polyp	Occult blood, iron deficiency, change in bowel habits, weight loss, or visible bleeding.	Colonoscopy with biopsy/polypectomy.	Stage-directed surgery/oncology; surveillance after resection or polypectomy.
IBD	Young adult or chronic relapsing diarrhea, pain, weight loss, urgency; bleeding more typical in UC.	Colonoscopy with biopsies plus inflammatory markers and stool studies.	Disease-specific anti-inflammatory or immunosuppressive therapy.
Hemorrhoids/fissure	Bright red blood on paper or in bowl; pain suggests fissure; hemorrhoids may itch or protrude.	Anoscopy/rectal exam, but do not stop evaluation if red flags exist.	Fiber, topical measures, banding/procedure if persistent; evaluate older or anemic patients for proximal disease.
Aortoenteric fistula	Prior aortic graft/aneurysm repair; herald bleed followed by catastrophic bleeding.	CTA and urgent vascular surgery.	Surgical emergency; do not delay for routine endoscopy if suspected.

Colorectal cancer: prevention, detection, and clinical patterns

Most colorectal cancers arise through a stepwise adenoma-carcinoma sequence in which normal epithelium becomes an adenomatous polyp, accumulates genetic alterations, develops dysplasia, and eventually invades through the muscularis mucosa. This progression usually takes years, which is why screening can prevent cancer rather than simply detect it. Colonoscopy is both diagnostic and therapeutic because premalignant adenomas can be removed during the same procedure. Stool-based screening detects occult bleeding or abnormal DNA but cannot remove lesions; a positive test must be followed by colonoscopy.

Average-risk screening begins at age 45. Earlier screening is required for high-risk groups, including a first-degree relative with colorectal cancer or advanced adenoma, hereditary polyposis syndromes, Lynch syndrome, inflammatory bowel disease involving the colon, prior abdominal or pelvic radiation, or previous advanced adenomas. For family-history screening, a common rule is to begin at age 40 or 10 years before the youngest affected first-degree relative, whichever comes first. The exact interval depends on the number, age, and degree of affected relatives and the findings at colonoscopy.

Screening option	Typical interval for average risk	Strength	Limitation
Colonoscopy	Every 10 years if normal	Visualizes entire colon; allows biopsy and polypectomy.	Requires bowel prep, sedation in many settings, procedural risk.
FIT	Every year	Detects human globin; more specific for lower GI bleeding than guaiac testing.	Requires annual adherence; positive test requires colonoscopy.
High-sensitivity gFOBT	Every year	Low cost and noninvasive.	Diet/medication false positives; less specific than FIT.
Stool DNA-FIT	Every 1-3 years depending on program	Detects DNA markers plus blood; higher sensitivity for some lesions.	More false positives than FIT; positive requires colonoscopy.
CT colonography	Every 5 years	Noninvasive structural exam.	Still requires bowel prep; abnormal findings require colonoscopy.
Flexible sigmoidoscopy	Every 5 years, or every 10 years plus annual FIT in some strategies	Examines distal colon where many lesions occur.	Misses proximal lesions unless combined with stool testing.

Symptoms of colorectal cancer are often late and insensitive. Right-sided cancers commonly cause occult bleeding and iron deficiency anemia because the cecum and ascending colon have a large caliber and liquid stool. Left-sided cancers more often cause altered bowel habits and obstructive symptoms because the descending/sigmoid colon is narrower and stool is more formed. Rectal cancers may cause tenesmus, urgency, incomplete evacuation, rectal bleeding, pelvic pain, or a palpable lesion. Any adult with unexplained iron deficiency anemia, positive stool-based test, persistent rectal bleeding, or alarm symptoms needs diagnostic evaluation rather than empiric hemorrhoid treatment alone.

High-yield distinction

Right-sided colon cancer: occult bleeding, iron deficiency anemia, fatigue, melena-like stool, vague pain. Left-sided colon cancer: hematochezia, constipation, narrowed stool caliber, obstruction. Rectal cancer: tenesmus, urgency, incomplete evacuation, visible blood, pelvic symptoms.

Colon cancer staging and treatment logic

Treatment depends on stage, location, surgical candidacy, mismatch repair status, metastatic pattern, and patient goals. Localized colon cancer is usually treated with surgical resection of the involved segment and regional lymphadenectomy. Rectal cancer often requires a different pathway because pelvic anatomy makes local recurrence more likely and radiation can be part of multimodality treatment. CEA is not a screening test. It can be checked before surgery for prognosis and followed after treatment as one recurrence marker, but it cannot replace imaging, colonoscopy, or clinical assessment.

Concept	Colon cancer	Rectal cancer
Curative local therapy	Segmental colectomy with lymph node evaluation.	Total mesorectal excision; often combined with radiation and/or chemotherapy depending on stage.
Radiation role	Not routinely used for typical colon cancer.	Commonly used for locally advanced rectal cancer to reduce pelvic recurrence.
Recurrence surveillance	History, exam, CEA when appropriate, colonoscopy, and CT-based surveillance for selected stages.	Similar principles, plus attention to pelvic recurrence and anastomotic/local disease.
Emergency presentation	Obstruction, perforation, bleeding, or metastatic disease.	Bleeding, obstruction, tenesmus, pain, and local invasion symptoms.

Colonic polyps and hereditary syndromes

A polyp is a gross description, not a diagnosis. Hyperplastic polyps are usually benign, especially when small and distal. Inflammatory pseudopolyps occur in inflammatory bowel disease and reflect prior mucosal injury rather than a neoplastic clone. Adenomas are premalignant epithelial neoplasms and include tubular, tubulovillous, and villous histology. Cancer risk rises with size, villous architecture, high-grade dysplasia, sessile morphology, and multiplicity. Serrated lesions, especially sessile serrated lesions in the proximal colon, can also progress to cancer through a pathway that differs from the classic adenoma-carcinoma sequence.

Polyp type	Risk profile	Important associations
Hyperplastic	Usually low malignant potential when small and distal.	Can be part of serrated polyposis when numerous or proximal.
Tubular adenoma	Most common adenoma; lower risk than villous if small and low grade.	Remove completely and set surveillance by size/number/histology.
Tubulovillous adenoma	Intermediate risk.	More concerning than tubular, less than villous.
Villous adenoma	Higher malignant potential; may cause secretory diarrhea and hypokalemia when large.	Large sessile rectosigmoid lesions are especially important.

Polyp type	Risk profile	Important associations
Sessile serrated lesion	Can be subtle, flat, proximal, and premalignant.	Associated with serrated pathway cancers.
Inflammatory pseudopolyp	Marker of prior inflammation rather than classic neoplastic risk.	Seen in ulcerative colitis and Crohn colitis.

Hereditary syndromes create recognizable clusters. Familial adenomatous polyposis causes hundreds to thousands of adenomas and near-inevitable colorectal cancer without colectomy, often by mid-adulthood. Gardner syndrome is an APC-associated variant with extracolonic manifestations such as osteomas, dental abnormalities, epidermoid cysts, and desmoid tumors. Turcot syndrome links colorectal polyposis with central nervous system tumors. Peutz-Jeghers syndrome causes hamartomatous polyps and mucocutaneous pigmentation; it increases risk of several cancers despite hamartomas having different malignant behavior than adenomas. Lynch syndrome is hereditary nonpolyposis colorectal cancer caused by mismatch repair defects; it increases colorectal cancer and extracolonic cancers such as endometrial, ovarian, gastric, small bowel, hepatobiliary, urinary tract, brain, and skin tumors.

Syndrome	Key mechanism/phenotype	Cancer concern
FAP	APC mutation; hundreds to thousands of adenomas; early onset.	Colorectal cancer risk approaches 100% without colectomy.
Gardner	FAP variant with osteomas, dental abnormalities, epidermoid cysts, desmoid tumors.	CRC plus extracolonic tumor burden.
Turcot	Polyposis with CNS tumors.	CRC and medulloblastoma/glioblastoma associations depending on genotype.
Peutz-Jeghers	Hamartomatous polyps plus mucocutaneous pigmentation.	GI, pancreatic, breast, gynecologic, and other cancers.
Lynch syndrome	Mismatch repair defect; few polyps but rapid progression.	CRC, endometrial, ovarian, gastric, urinary tract, and other cancers.

Diverticulosis and diverticulitis

Diverticula are acquired outpouchings of mucosa and submucosa through weak points in the colonic muscular wall, often where vasa recta penetrate. Diverticulosis is common with aging and is most common in the sigmoid colon, though right-sided diverticula occur. The traditional association with low-fiber diets is clinically useful because constipation and segmentation can raise intraluminal pressure. Most patients are asymptomatic. Two major complications dominate internal medicine: painless diverticular hemorrhage and diverticulitis.

Diverticular bleeding occurs when the vasa recta associated with a diverticulum is injured. It is classically painless and can be brisk. It often stops spontaneously, but recurrent bleeding can occur. Diverticulitis occurs when microperforation and inflammation develop around a diverticulum. The usual presentation is left lower quadrant pain, fever, leukocytosis, nausea, constipation or diarrhea, and focal tenderness. A phlegmon, abscess, fistula, obstruction, free perforation, or generalized peritonitis transforms simple inflammation into complicated disease.

Feature	Diverticulosis	Uncomplicated diverticulitis	Complicated diverticulitis
Pathology	Diverticula without acute inflammation.	Localized inflammation/microperforation without abscess, perforation, obstruction, or fistula.	Abscess, free perforation, obstruction, fistula, sepsis, or peritonitis.
Symptoms	Usually none; possible painless hematochezia.	LLQ pain, fever, leukocytosis, bowel habit change.	Severe pain, guarding, persistent fever, sepsis, mass, urinary/pneumaturia symptoms, obstruction.
Preferred diagnosis	Usually incidental; colonoscopy may find diverticula.	CT abdomen/pelvis with contrast in many first or uncertain cases.	CT to define abscess/perforation; surgical consultation.

Feature	Diverticulosis	Uncomplicated diverticulitis	Complicated diverticulitis
Treatment	Fiber and constipation prevention; manage bleeding if present.	Selective antibiotics in mild immunocompetent patients; antibiotics for higher-risk patients; clear liquids then advance.	Antibiotics, drainage if abscess is large, surgery if perforation/peritonitis/failed management.

Modern management anchor

Antibiotics are no longer automatic for every mild, uncomplicated diverticulitis episode in an immunocompetent patient. They are used for complicated disease, systemic inflammation, immunocompromise, significant comorbidity, refractory symptoms, vomiting, high inflammatory markers, or concern for abscess/perforation.

Colonoscopy is avoided during acute diverticulitis because insufflation and instrumentation can worsen perforation risk. After recovery, colonoscopy is often considered if the patient has not had recent high-quality colon evaluation, especially after complicated diverticulitis or when cancer could mimic the CT findings. Recurrent uncomplicated episodes are not automatically a surgery indication; elective surgery is individualized based on severity, complications, immune status, quality of life, and patient preference.

Angiodysplasia and vascular ectasia

Angiodysplasia refers to dilated, tortuous, thin-walled vessels in the mucosa and submucosa, often in the right colon. It is a common cause of lower GI bleeding in older adults and can produce chronic iron deficiency anemia, intermittent occult bleeding, or overt painless hematochezia. Bleeding may stop spontaneously and recur unpredictably. Associations include age, chronic kidney disease, von Willebrand disease, and aortic stenosis. The aortic stenosis association is often discussed as Heyde syndrome, in which shear stress may contribute to acquired von Willebrand factor dysfunction.

Colonoscopy can diagnose and treat colonic angiodysplasia. Endoscopic argon plasma coagulation or other coagulative techniques are common when a bleeding or high-risk lesion is found. If bleeding is active and severe, CT angiography can help localize the site, and angiography can embolize. Recurrent obscure bleeding after negative EGD and colonoscopy may require capsule endoscopy, deep enteroscopy, or targeted imaging to evaluate the small bowel.

Ischemic colitis and acute mesenteric ischemia

Intestinal ischemia is a spectrum. Ischemic colitis usually reflects transient hypoperfusion or localized vascular compromise of the colon, often at watershed areas such as the splenic flexure or rectosigmoid junction. It classically presents with crampy abdominal pain followed by bloody diarrhea. Acute mesenteric ischemia is usually more catastrophic because interruption of blood flow to the small bowel or proximal colon can cause rapid transmural infarction. Pain out of proportion to the abdominal exam is the classic clue, but this clue can disappear late when peritonitis develops.

Type	Mechanism	Typical setting	Diagnostic clue	Treatment principle
Arterial embolism	Sudden occlusion, often SMA.	Atrial fibrillation, recent MI, valvular disease.	Abrupt severe pain, little early tenderness.	CTA, anticoagulation, revascularization/embolectomy, surgery if necrosis.
Arterial thrombosis	Plaque rupture or thrombosis on chronic atherosclerosis.	CAD, PAD, prior postprandial pain/weight loss.	Severe pain, often with chronic mesenteric ischemia history.	Revascularization and bowel viability assessment.
Nonocclusive mesenteric ischemia	Low-flow vasoconstrictive state without fixed clot.	Shock, sepsis, heart failure, vasopressors, dialysis.	Critically ill patient with acidosis/distention/pain.	Restore perfusion, reduce vasoconstriction when possible, papaverine in selected cases.

Type	Mechanism	Typical setting	Diagnostic clue	Treatment principle
Mesenteric venous thrombosis	Venous outflow obstruction causing edema and ischemia.	Hypercoagulability, cancer, pancreatitis, portal HTN, OCPs.	More subacute pain; CT venous findings.	Anticoagulation; surgery if infarction/peritonitis.

Serum lactate is important but not reassuring when normal early. Lactate elevation, metabolic acidosis, leukocytosis, hemoconcentration, elevated D-dimer, and shock support severity but cannot exclude ischemia. CT angiography is the practical first-line imaging test for suspected acute mesenteric ischemia in many modern pathways because it can identify arterial occlusion, venous thrombosis, bowel wall changes, pneumatosis, portal venous gas, and alternative diagnoses. Diagnostic delay is dangerous: mortality can be high, and when bowel infarction occurs, death risk rises substantially. Peritonitis, perforation, necrosis, or clinical deterioration requires urgent surgery.

Severe abdominal pain out of proportion to exam

Ask about atrial fibrillation, CAD/PAD, low-flow shock, vasopressors, hypercoagulability, and prior postprandial pain.

THEN

Immediate actions

NPO, IV fluids, broad-spectrum antibiotics if ischemia suspected, labs including lactate, type and screen, early surgery/vascular consultation.

THEN

CTA abdomen/pelvis

Look for SMA/IMA/cealic occlusion, venous thrombosis, bowel wall hypoenhancement, pneumatosis, portal venous gas, free air.

THEN

Treatment branch

Revascularize embolic/thrombotic arterial disease, anticoagulate venous thrombosis, reverse low-flow state in nonocclusive ischemia, resect necrotic bowel.

Chronic mesenteric ischemia

Chronic mesenteric ischemia is abdominal angina. Atherosclerotic narrowing of the celiac, superior mesenteric, and/or inferior mesenteric arteries limits postprandial blood flow. Patients develop dull, cramping epigastric or periumbilical pain 15-60 minutes after meals, fear of eating, and weight loss. Because at least two major vessels are often affected before symptoms become prominent, the disease implies systemic atherosclerosis. Diagnosis is made with vascular imaging such as CTA, MRA, duplex ultrasound in selected settings, or angiography. Definitive therapy is revascularization, either endovascular or surgical depending on anatomy and patient risk.

Ogilvie syndrome, ileus, and colonic pseudo-obstruction

Ogilvie syndrome, or acute colonic pseudo-obstruction, produces signs and radiographic evidence of large bowel obstruction without a mechanical blockage. It is common in hospitalized or postoperative patients and is associated with severe medical illness, sepsis, trauma, immobility, electrolyte abnormalities, neurologic disease, and medications such as opioids, anticholinergics, and psychotropics. The colon becomes progressively dilated, often with prominent cecal dilation. The danger is not the label; the danger is wall tension, ischemia, and perforation.

Disorder	Mechanical blockage?	Gas pattern	Common causes	Management
Mechanical large bowel obstruction	Yes	Dilated colon proximal to obstruction; little distal gas if complete.	Cancer, volvulus, stricture, hernia.	CT, decompression when appropriate, surgery/endoscopic therapy depending cause.

Disorder	Mechanical blockage?	Gas pattern	Common causes	Management
Paralytic ileus	No	Diffuse gas throughout small bowel and colon, including rectum.	Postoperative state, opioids, anticholinergics, hypokalemia, peritonitis, shock.	NPO, IV fluids, electrolyte correction, stop offending drugs, mobilization, NG if vomiting.
Ogilvie syndrome	No fixed lesion, but colon behaves obstructed.	Marked colonic dilation, often cecum/right colon.	Hospitalization, surgery, trauma, sepsis, neurologic disease, medications.	Supportive care, correct triggers; neostigmine or colonoscopic decompression if severe/persistent; surgery if perforation/ischemia.

Conservative management includes bowel rest, IV fluids, correction of hypokalemia and hypomagnesemia, stopping opioids/anticholinergics when possible, mobilization, rectal tube or gentle decompression in selected patients, and serial abdominal exams and imaging. If cecal diameter approaches 10-12 cm, dilation persists, or symptoms worsen, escalation is needed. Neostigmine can be effective but requires monitoring because of bradycardia and bronchospasm risk. Colonoscopic decompression is used when pharmacologic therapy fails or is contraindicated. Peritonitis, ischemia, or perforation requires surgery.

Colonic volvulus

Volvulus is twisting of bowel around its mesenteric attachment, creating a closed-loop obstruction that can compromise vascular supply. Sigmoid volvulus is most common and is associated with older age, chronic constipation, institutionalization, neuropsychiatric disease, and a redundant sigmoid colon. Cecal volvulus is less common and often occurs in younger patients with congenital malfixation or prior abdominal surgery. Both can produce obstruction, ischemia, perforation, and sepsis.

Feature	Sigmoid volvulus	Cecal volvulus
Frequency	Most common colonic volvulus.	Second most common.
Risk factors	Older age, chronic constipation, institutionalization, neurologic or psychiatric disease.	Younger patients, congenital malfixation, prior surgery, pregnancy, adhesions.
Imaging clue	Coffee-bean or omega-loop dilated sigmoid, often apex toward RUQ.	Coffee-bean or dilated cecum often displaced to LUQ or mid-abdomen.
Initial treatment if no ischemia/perforation	Flexible sigmoidoscopic detorsion with rectal tube; elective sigmoid colectomy often recommended because recurrence is high.	Endoscopic decompression is less successful; surgery is usually required.
Emergency surgery triggers	Peritonitis, perforation, ischemia, failed decompression, unstable patient.	Usually surgical management; urgent if ischemia or perforation suspected.

Large bowel obstruction and emergency thresholds

Large bowel obstruction is most commonly caused by colorectal cancer, volvulus, diverticular stricture, fecal impaction, or inflammatory stricture. Compared with small bowel obstruction, large bowel obstruction may cause more marked abdominal distention and later vomiting. The ileocecal valve matters: if competent, a closed-loop obstruction can form between the obstructing lesion and the valve, causing rapid cecal dilation and perforation risk. The cecum is most vulnerable because wall tension rises with radius.

Red flags include fever, tachycardia, hypotension, continuous severe pain, peritoneal signs, leukocytosis, metabolic acidosis, elevated lactate, pneumatosis, portal venous gas, free air, and cecal dilation near or above 10-12 cm. Management depends on cause and stability. Cancer obstruction may require stenting as a bridge to surgery in selected left-sided lesions, urgent resection, or diversion. Volvulus requires detorsion or surgery. Pseudo-obstruction requires exclusion of mechanical obstruction and decompression when danger thresholds are approached.

Integrated diagnostic algorithms

Algorithm 1: Hematochezia

Bright red blood per rectum

Assess hemodynamics, stool volume, anticoagulants, liver disease, cardiac disease, age, prior colonoscopy, and abdominal pain.

THEN

Unstable or brisk?

Resuscitate. Consider upper GI source. EGD if suspicion for upper bleeding; CTA if active severe lower bleeding.

THEN

Stable

Rectal exam/anoscopy if obvious anorectal bleeding, but age >45, anemia, weight loss, family history, or persistent bleeding still needs colon evaluation.

THEN

Colonoscopy after prep

Find and treat diverticular bleeding, angiodysplasia, cancer, IBD, post-polypectomy bleeding, ischemic colitis, or hemorrhoids.

Algorithm 2: Left lower quadrant pain

LLQ pain +/- fever/leukocytosis

Consider diverticulitis, ischemic colitis, IBD, colon cancer obstruction, gynecologic/urinary causes, constipation, and infectious colitis.

THEN

Peritonitis, sepsis, severe pain, immunocompromise?

CT with contrast, IV antibiotics, surgery consultation; avoid colonoscopy during acute diverticulitis.

THEN

Mild uncomplicated picture

CT in first or uncertain episode; selective antibiotics depending risk; clear liquids then advance; follow-up colon evaluation when indicated.

Algorithm 3: Abdominal pain out of proportion

Pain out of proportion to exam

Think acute mesenteric ischemia until proven otherwise, especially with atrial fibrillation, atherosclerosis, low-flow shock, or hypercoagulability.

THEN

Do not wait for peritonitis

Labs may be nonspecific early. Lactate can be late. Begin resuscitation and consult surgery/vascular early.

THEN

CTA abdomen/pelvis

Treat by mechanism: embolus, thrombosis, nonocclusive low-flow, or venous thrombosis.

High-yield numeric anchors

Number	Meaning
45 years	Average-risk colorectal cancer screening begins.
75 years	Routine screening generally continues through this age if healthy enough and previously unscreened/appropriately screened.
76-85 years	Individualize CRC screening based on health, prior screening, and preferences.
>85 years	Routine CRC screening is generally stopped.
10 years	Typical interval after normal average-risk colonoscopy.
Age 40 or 10 years before youngest affected relative	Common family-history starting point for CRC screening when a first-degree relative has CRC/advanced adenoma.
5-10%	Approximate proportion of hematochezia cases that may originate from a brisk upper GI bleed in some clinical series.
4-8 hours	Common interval for hemoglobin trending during active GI bleeding until stable.
7 g/dL	Common restrictive transfusion threshold in stable GI bleeding; higher targets may be used in shock or significant cardiopulmonary disease.
10-12 cm cecum	Danger range for severe colonic dilation/pseudo-obstruction; perforation risk rises with size and duration.
>24 hours ongoing severe bleeding	One historical surgical concern when bleeding continues despite intervention.
5 units in 4-6 hours	Large transfusion requirement suggests severe ongoing bleeding and may prompt surgery/advanced intervention.
Pain 15-60 minutes after meals	Typical chronic mesenteric ischemia/abdominal angina timing.
High mortality if infarction occurs	Acute mesenteric ischemia becomes far more lethal after transmural bowel necrosis, sepsis, and shock.

Common pitfalls

- Calling red blood from the rectum lower GI bleeding without assessing shock. Brisk upper GI bleeding can present as hematochezia.
- Treating presumed hemorrhoids in an older patient with iron deficiency anemia, weight loss, positive stool test, or persistent bleeding without colon evaluation.
- Using CEA as a screening test. CEA is used mainly for prognosis and surveillance in selected patients after diagnosis.
- Performing colonoscopy during acute diverticulitis. Colonoscopy is delayed until recovery when colon evaluation is needed.
- Withholding concern for mesenteric ischemia because lactate is normal early. Normal lactate does not exclude early ischemia.
- Missing Ogilvie syndrome in a postoperative or medically ill patient with progressive colonic dilation and no mechanical lesion.
- Assuming successful sigmoid detorsion cures sigmoid volvulus. Recurrence is common without definitive management.
- Forgetting medication causes of bleeding and ileus: NSAIDs, aspirin, clopidogrel, anticoagulants, opioids, anticholinergics, and psychotropics.

Practice questions

Question: A 68-year-old man has sudden painless large-volume hematochezia. Blood pressure is 98/60, pulse 116. He takes apixaban for atrial fibrillation. What is the first management priority?

Answer: Resuscitation and stabilization: assess ABCs, place large-bore IV access, type/cross, labs, transfuse as needed, hold/reverse anticoagulation if severe/life-threatening, then localize bleeding. Do not start with outpatient colonoscopy planning.

Question: A 57-year-old woman has fatigue and microcytic anemia. FIT is positive. She has no abdominal pain. What must happen next?

Answer: Diagnostic colonoscopy. A positive stool-based screen is not a diagnosis and not a completed screening pathway.

Question: A 74-year-old with atrial fibrillation develops abrupt severe periumbilical pain but has a soft abdomen. Lactate is mildly elevated. What diagnosis is most dangerous to miss?

Answer: Acute mesenteric ischemia from arterial embolism. Pain out of proportion to exam should prompt CTA and early surgical/vascular involvement.

Question: A hospitalized postoperative patient develops marked abdominal distention. CT shows diffuse colonic dilation without a transition point or mass. Cecum is 11 cm. What syndrome is likely?

Answer: Ogilvie syndrome, or acute colonic pseudo-obstruction. Supportive care and correction of triggers are initial steps, but severe/persistent dilation may require neostigmine or colonoscopic decompression.

Question: An older institutionalized patient has abdominal distention, constipation, and a coffee-bean shaped loop on abdominal radiograph. What is the likely diagnosis and initial therapy if no peritonitis?

Answer: Sigmoid volvulus. Flexible sigmoidoscopic detorsion and rectal tube placement are typical initial therapy when there is no ischemia/perforation, followed by consideration of definitive surgery because recurrence is common.

Question: A patient recovering from left lower quadrant pain and CT-proven complicated diverticulitis asks why colonoscopy was not done during admission.

Answer: Colonoscopy is avoided during acute diverticulitis because insufflation and instrumentation may worsen perforation risk. Colon evaluation is considered after recovery when indicated to exclude malignancy or other pathology.

Question: A patient with aortic stenosis and CKD has recurrent painless lower GI bleeding. Colonoscopy shows flat vascular ectasias in the right colon. What is the concept?

Answer: Angiodysplasia, sometimes associated with aortic stenosis/Heyde syndrome. Endoscopic coagulation can treat culprit lesions, but recurrence is possible.

Question: A patient has postprandial abdominal pain, food fear, and weight loss with diffuse atherosclerosis. What condition is suggested?

Answer: Chronic mesenteric ischemia, also called abdominal angina. Vascular imaging and revascularization are central concepts.

Graph-RAG extraction map

The following concept map is designed to support node and edge generation. Nodes should include diseases, symptoms, tests, thresholds, treatments, complications, and risk factors. Edges should specify direction and rationale, such as "causes," "suggests," "diagnosed by," "treated with," "contraindicated during," or "complicated by."

Source node	Relationship	Target node	Rationale
Colorectal cancer	arises from	adenomatous polyp	Most CRC follows an adenoma-carcinoma sequence over years.
FIT positive	requires	diagnostic colonoscopy	Stool tests detect risk; colonoscopy confirms and treats lesions.

Source node	Relationship	Target node	Rationale
Right colon cancer	presents with	iron deficiency anemia	Wider lumen allows occult bleeding before obstruction.
Left colon cancer	presents with	change in bowel habits	Narrower lumen and formed stool promote obstructive symptoms.
Diverticulosis	complicated by	painless hematochezia	Vasa recta injury can cause brisk bleeding.
Diverticulitis	diagnosed by	CT abdomen/pelvis	CT defines inflammation, abscess, perforation, and mimics.
Acute diverticulitis	contraindicates	colonoscopy during acute attack	Instrumentation can increase perforation risk.
Angiodysplasia	causes	recurrent lower GI bleeding	Fragile mucosal/submucosal ectatic vessels bleed intermittently.
Acute mesenteric ischemia	suggested by	pain out of proportion	Early ischemia causes severe visceral pain before peritoneal findings.
Atrial fibrillation	predisposes to	arterial embolic mesenteric ischemia	Embolus can occlude SMA branches.
Ogilvie syndrome	causes	colonic dilation without obstruction	Autonomic dysregulation impairs colonic motility.
Cecal diameter 10-12 cm	increases risk of	perforation	Wall tension rises with increasing radius.
Sigmoid volvulus	treated by	endoscopic detorsion	Untwisting decompresses bowel when no ischemia/perforation.
Cecal volvulus	treated by	surgery	Endoscopic decompression is less successful and ischemia risk is substantial.

Selected guideline cross-checks used during drafting

Current clinical anchors were checked against major published guidance and consensus resources on colorectal cancer screening, acute lower GI bleeding, diverticulitis, and acute mesenteric ischemia. This list is included to document validation targets, not to imply that the chapter reproduces guideline text.

- USPSTF colorectal cancer screening recommendation for average-risk adults, including initiation at age 45 and individualized screening in older adults.
- American College of Gastroenterology guidance for acute lower gastrointestinal bleeding, including stabilization first, colonoscopy after preparation in stable patients, and CTA/angiography roles for active severe bleeding.
- American Gastroenterological Association clinical practice update on diverticulitis, including selective antibiotic use in mild uncomplicated disease and CT-based confirmation in appropriate patients.
- World Society of Emergency Surgery updated guidance for acute mesenteric ischemia, including early suspicion, CTA, resuscitation, antibiotics, anticoagulation in selected cases, revascularization, and surgery for necrotic bowel.